

Vitaly Chait

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PROFESSIONAL SUMMARY

Software engineer with 10 years of experience building high-throughput backend services, distributed systems, and mission-critical infrastructure across the US, Israel, and China. Demonstrated track record integrating large-scale distributed frameworks, cutting integration cycles by >50%, and scaling automated test coverage by over 1,000%. Brings deep hands-on expertise in Python, Java, C++, REST APIs, WebSockets, and AWS to build and scale backend platforms.

TECHNICAL SKILLS

Backend & Cloud: REST APIs, Microservices, Flask, FastAPI, WebSockets, AWS EC2 & S3, Distributed Systems

Data Infrastructure: Apache Spark, Presto, ClickHouse, MySQL, ElasticSearch, InfluxDB, scikit-learn, OpenCV

DevOps & Systems: Linux, Docker, Git, Jenkins CI/CD, Debian Releases, Grafana, Prometheus Monitoring

Languages: Python, Java, C, C++, SQL, Bash, JavaScript, TypeScript, MATLAB

AI: RAG, MCP, Prompt & Tool Proficiency, AI Governance and Security

PROFESSIONAL EXPERIENCE

Software Engineer — NeuroBlade (acquired by AWS)

Jan 2023 – Sep 2025

Tel Aviv, Israel

- Led cross-team integration of components spanning Apache Spark, Presto, ClickHouse, InfluxDB, ElasticSearch, and AWS EC2 and S3, increasing test coverage by >1,000% and cutting integration time by over 50%.
- As part of the acceleration team, integrated data accelerator with distributed query engines (Presto, Spark), improving relative performance from 0.8x to 4x versus baseline CPU execution; wrote software for embedded hardware kernel components.
- Built robust Jenkins CI/CD and Debian release pipelines; configured Kubernetes clusters and Grafana, Prometheus, MySQL, and InfluxDB monitoring.
- Designed C-based diagnostic infrastructure across PCIe, NVMe, network switches, and FPGAs, reducing development bottlenecks from months to weeks.

Software Engineer — Self-employed / Independent Contractor

Jul 2021 – Jan 2023

Beijing, China

- Designed and deployed production backend services and REST APIs for SoC-based products using Flask, Python, and SQL; applied TFLite supervised ML models with OpenCV for image and motion detection.
- Helped create low-cost software-hardware interfaces for embedded products.

System Engineer — Orbotech (Acquired by KLA)

Aug 2018 – Jul 2021

Tel Aviv, Israel

- Led system engineering for Micro-LED deployments across sensor fusion, optics, motion algorithms, and software; delivered <0.5 micron defect-detection accuracy at 60% gross margin.
- Researched optical imaging systems for display technologies, prototyping in Python and MATLAB; collaborated with engineering teams and customers in Cupertino, Shanghai, and Beijing.
- Outstanding Employee, FPD R&D; (Jun 2019) — selected from 200+ employees.

Resident Engineer — Orbotech (Acquired by KLA)

Aug 2017 – Jul 2018

Beijing, China

- Liaised between international R&D; and field engineering teams during customer deployments, streamlining global rollout communication.

SELECTED PROJECTS

PolyGains (Co-Developer) — polygains.com

- Built backend and data streaming pipelines to monitor decentralized prediction market order flow, wallets, and trade signals.
- Engineered real-time transaction ingestion and wallet clustering algorithms; won the Brandeis Ain Family Start-Up Award.

EDUCATION & CREDENTIALS

MS, Business Analytics, Brandeis University

Dec 2026

- Graduate Researcher under Prof. Shubhramshu Shekhar; Teaching Assistant for BUS 244A 'Generative AI Foundations for Business'.
- Build and evaluate applied AI workflows and service integrations using Python, PyTorch, and ML libraries.
- Develop evaluation pipelines and automated workflows for model benchmarking and API integration testing; industry capstone: multi-asset meta-labeling portfolio pipeline (State Street Global Advisors).

BS, Computer Science, The Open University of Israel

2022

LEADERSHIP & SERVICE

Mentored at-risk youth through sailing programs and university scholars through Watoobi. Served 3 years as a military instructor for senior leadership (2014–2017).